TEACHING WALLAH

Teaching Exam

Subject – Computer Science

Topic – Computer Network

Lecture No.- 1

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Topics to be Covered

Topic

1:- Fundamental of Computer Network
2:- Advantage of Computer Network
3:- Disadvantage of Computer Network
4:- Components of Computer Network
5:- Network Device

What is Network ?

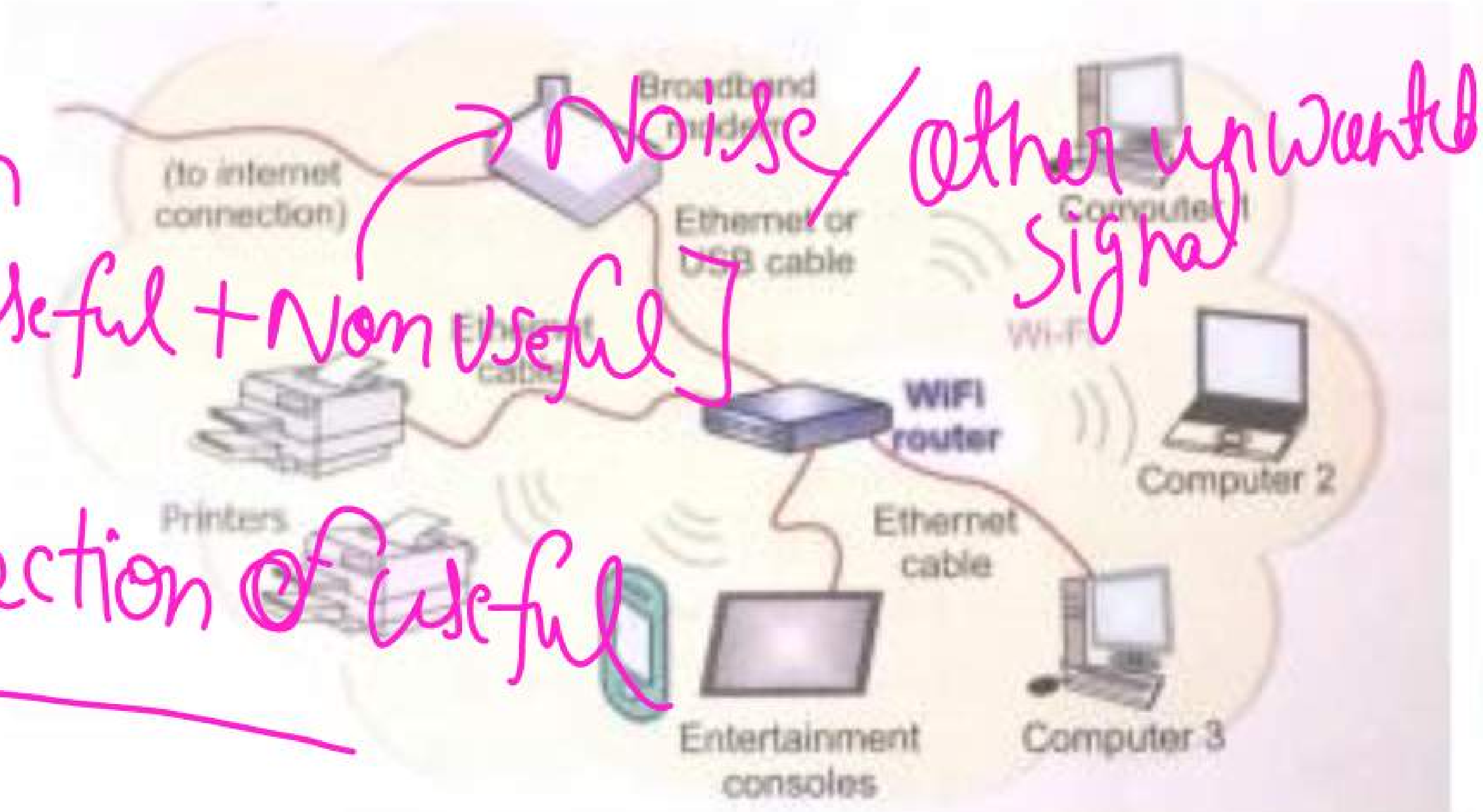
Computer Network

When a number of Computer or Computing Devices connected to one other in order to exchange information or share resources, from a computer network

Example:- In home you can Connect your Smartphone, your laptop with your Smart TV.

data is a collection of Raw message [useful + Non useful] → Noise/other unwanted signal

Information is a collection of useful data only



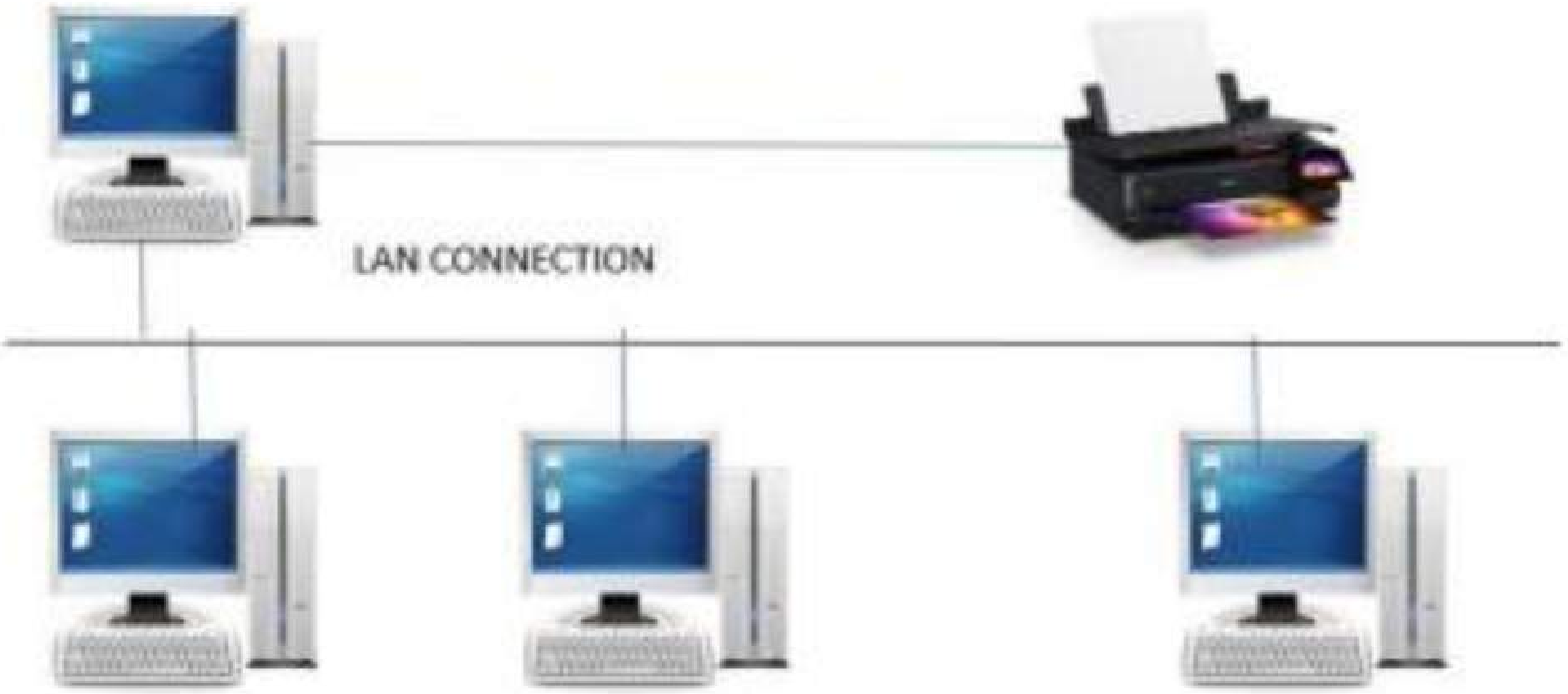
Topic : Computer Network

wired

- **Sharing devices such as printers and scanners**: Multiple systems can access the same hardware, reducing the need for duplicate devices and lowering costs.
- **Sharing Data**: Teams can work on shared documents, applications or systems, which boosts efficiency.
- **Communicating using web, email, video and instant messaging**: Networks enable both real-time and delayed communication. Users can access information, send messages, participate in video calls and chat.
- **Data management**: Networks allow organizations to store data in a central or distributed location, making it easier to manage, secure and back up critical information.
- **Remote access** : Users can log into computers, servers or cloud platforms from different locations, supporting remote work and 24/7 access.

Topic : Advantage of Computer Network

- **Cost Effective**:- Many Hardware Devices , Such as Printers, Scanners and Modems, can also be Shared by all the Computers on a Network, Thus it save a Significant amount of Money.



Topic : Advantage of Computer Network

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- **BACK UP**:- Network faces Technical issues or hardware failure , the work can be allocated to another computer with in network.

- **Easy Access to Remote Database:-** one can instantly book railway or airlines via Internet (Global network)

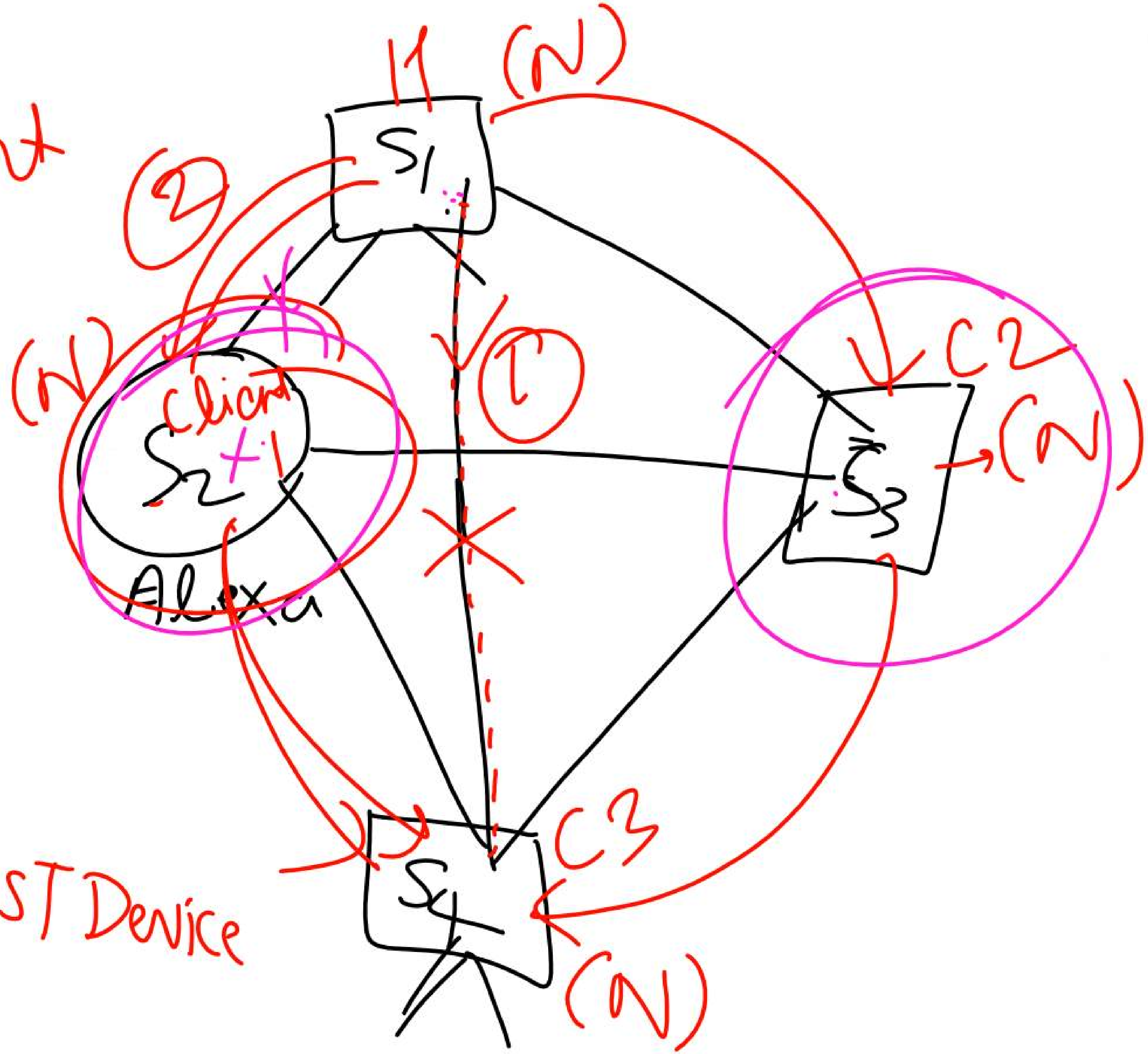
- **Real time Delivery:-** One can instantly Share Data and Information across the Globe.

HOST/Client

⇒ Node

S1 → S4

Communication initiate(Start) → HOST Device



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Topic:- Disadvantage of Computer Network

- The Systems are more Sophisticated and Complex to Run. This can add to costs and you may need Specialist Staff to Run the Network.

- If networks are Badly Managed Services can become Unusable and Productivity falls.

- File Security is more important especially if connected to WAN network. Ex:- Protection from Viruses.

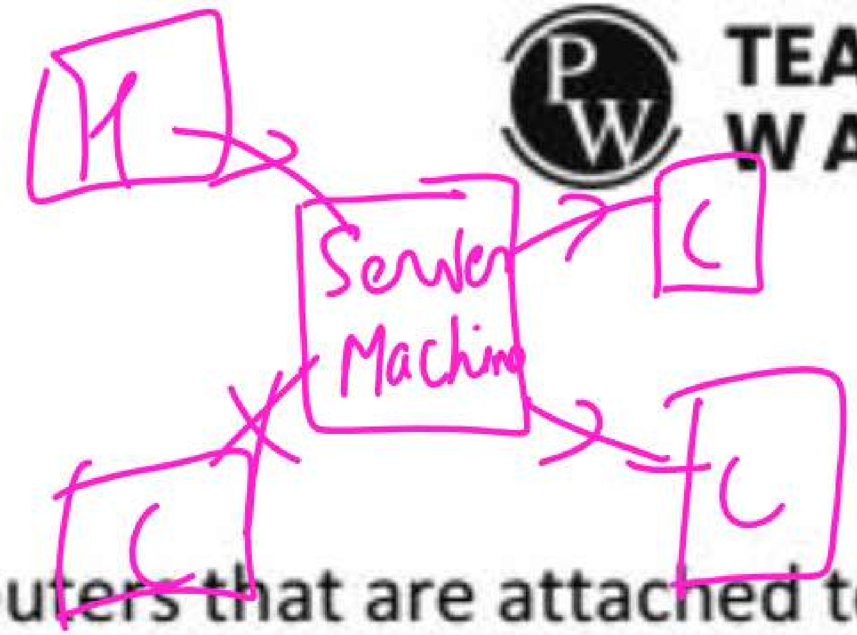
cmd

Ping google.com -t

13 ms

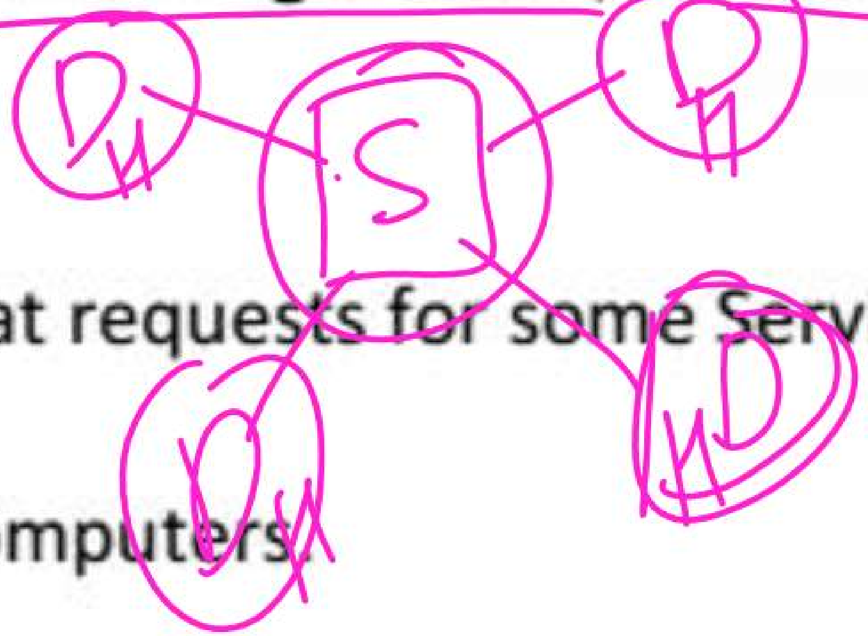
14 ms

Topic- Components of a Computer Network



- **Host / Nodes:-** The Term Host or Node Refers to the Computers that are attached to a Network are Seeking to Share the resources of the network.

- **Servers**:- A Servers Facilitates networking tasks like sharing of data , resource Sharing , Communication among hosts etc.



- **Clients:-** A Client Computer is a host Computer that requests for some Services from a Server.

Node

A Server Computer Serves the request of Client Computers.

Topic- Network Devices

Common Types of Networking Devices and Their Uses

Network devices work as a mediator between two devices for transmission of data, and thus play a very important role in the functioning of a computer network. Below are some common network devices used in modern networks:

- Access Point
- Modems
- Firewalls
- Repeater
- Hub
- Bridge
- Switch
- Routers
- Gateway
- Brouter
- NIC

Topic- Network Device- NIC CARD

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- **Network Hardware**
- **NIC CARD:-** It Stands for Network Interface Card (Hardware) attached to a host so as to establish network Connections.
- Every NIC Card has a unique physical address called MAC address(Media Access Control Address) which is a 6-byte(48 bits) addressed assigned by the NIC Manufacturer.

  It is also called Network Interface Unit (NIU) or Terminal Point(TAP)

NIC or network interface card is a network adapter that is used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique ID that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and the router or modem. NIC is a layer 2 device which means that it works on both the physical and data link layers of the network model.

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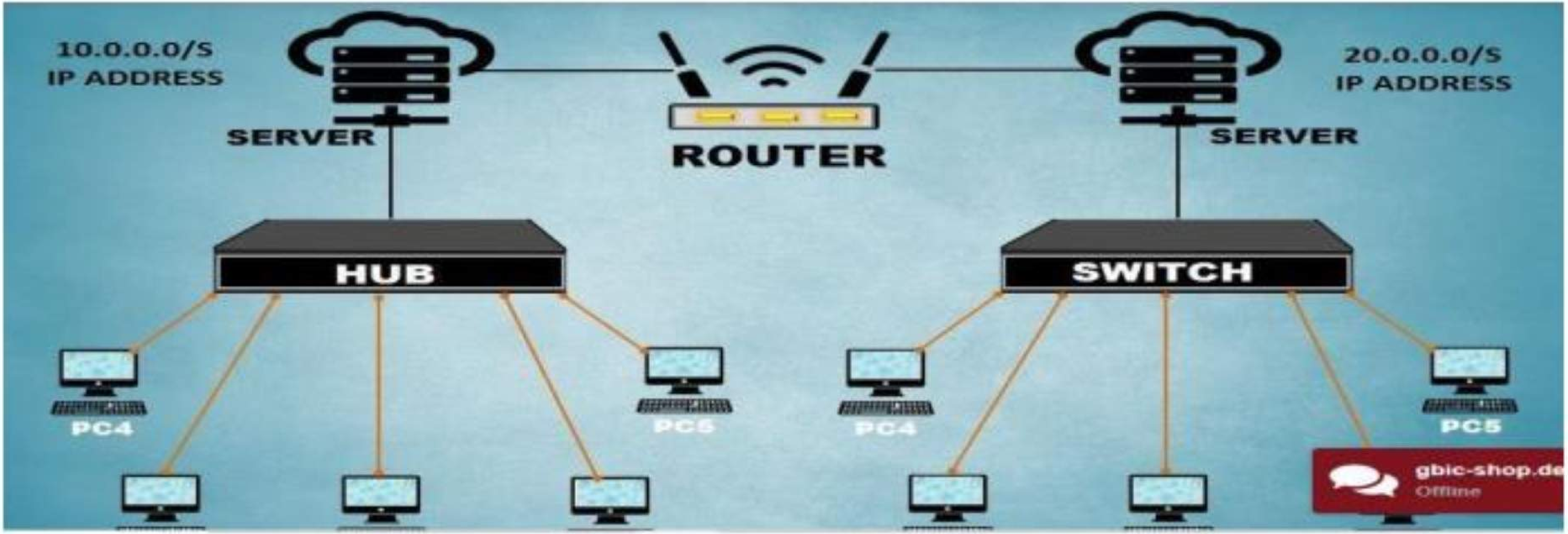
Topic- Network Device - HUB

HUB & SWITCH



Topic- Network Device - HUB

Hub

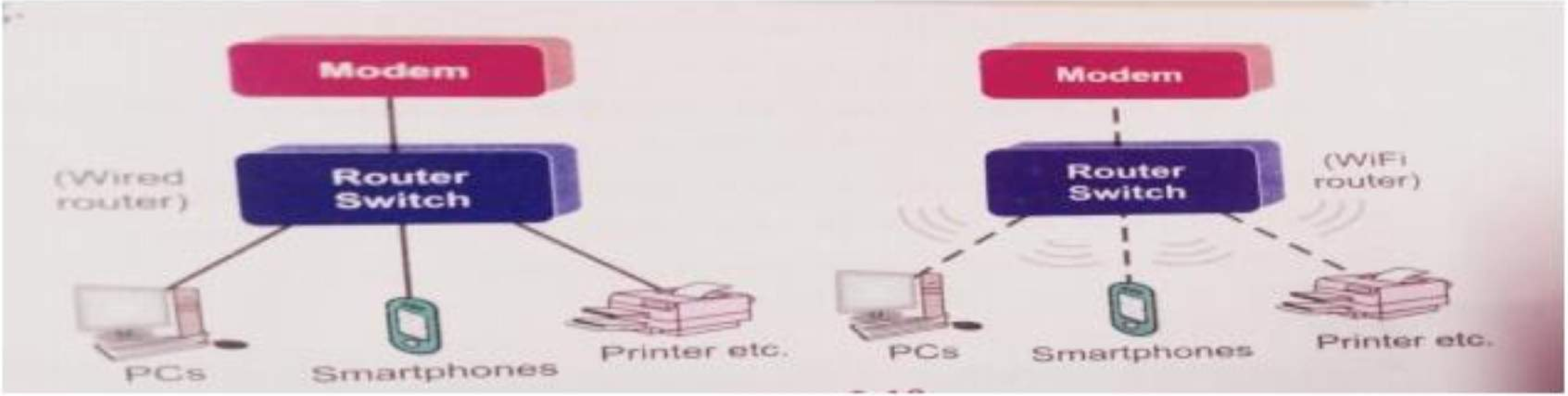
A hub is a multiport repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

Switch

A switch is a multiport bridge with a buffer designed that can boost its efficiency(a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only. In other words, the switch divides the collision domain of hosts, but the broadcast domain remains the same.

Topic- Network Device - ROUTER

Router:- The Router is responsible or Forwarding data from one Network to different network.



Router

A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.

Topic- Network Device – Access Point, Modems

Access Point

An access point in networking is a device that allows wireless devices, like smartphones and laptops, to connect to a wired network. It creates a Wi-Fi network that lets wireless devices communicate with the internet or other devices on the network. Access points are used to extend the range of a network or provide Wi-Fi in areas that do not have it. They are commonly found in homes, offices, and public places to provide wireless internet access.

Modems

Modem is also known as modulator/demodulator is a network device that is used to convert digital signal into analog signals of different frequencies and transmits these signals to a modem at the receiving location. These converted signals can be transmitted over the cable systems, telephone lines, and other communication mediums. A modem is also used to convert an analog signal back into digital signal. Modems are generally used to access the internet by customers of an Internet Service Provider (ISP).

Topic- Network Device – Firewall , Repeater

Firewalls

A firewall is a network security device that monitors and controls the flow of data between your computer or network and the internet. It acts as a barrier, blocking unauthorized access while allowing trusted data to pass through. Firewalls help protect your network from hackers, viruses, and other online threats by filtering traffic based on security rules. Firewalls can be physical devices (hardware), programs (software), or even cloud-based services, which can be offered as SaaS, through public clouds, or private virtual clouds.

Repeater

A repeater operates at the physical layer. Its main function is to amplify (i.e., regenerate) the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network. When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting following the original strength. It is a 2-port device.

Topic- Network Device – GATEWAY

What is Gateway?

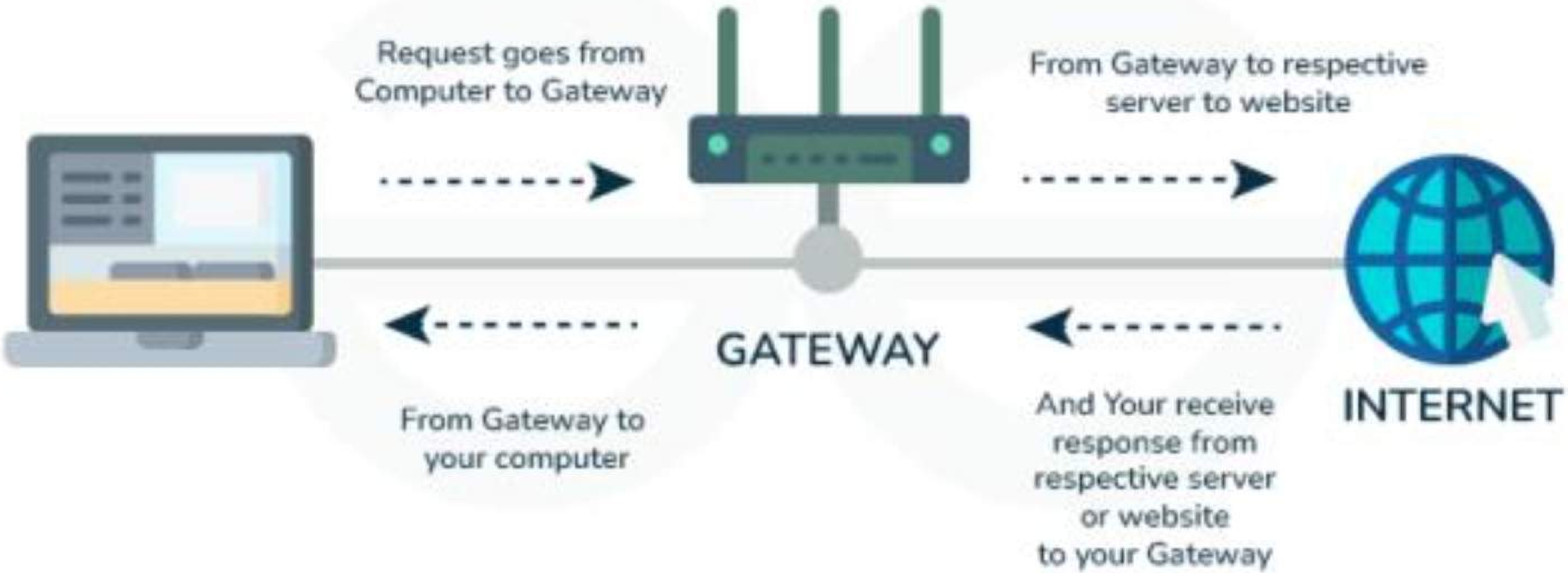
A gateway is a connecting point of any network that helps it to connect with different networks. The gateway monitors and controls all the incoming and outgoing traffic of the network. Suppose there are two different networks and they want to communicate with each other, so they need to set up a path between them. Now that path will be made between gateways of those different networks. Gateways are also known as protocol converters because they help to convert protocol supported by traffic of the different networks into that are supported by this network. Because of that, it makes smooth communication between two different networks.

How does Gateway Work?

Gateway has a simple working methodology of five steps:

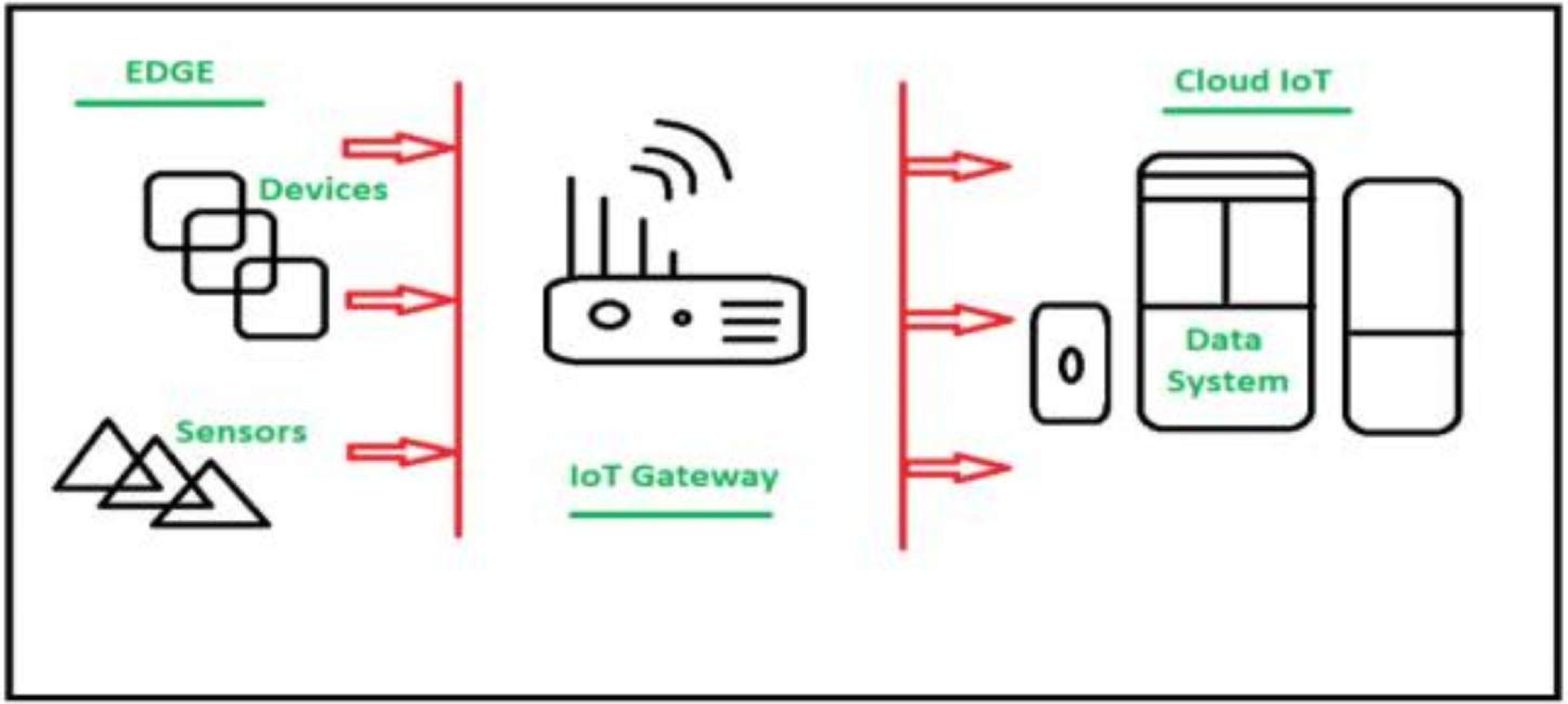
- **Step 1:** It gets data from the network
- **Step 2:** It intercepts and analyzes the received data.
- **Step 3:** It routes the data to the destination address.
- **Step 4:** It converts the received data to make that compatible with the receiver network.
- **Step 5:** It sends the final data inside the network.

Topic- Network Device – GATEWAY



Topic- Network Device – GATEWAY

Gateway is a device that connects different networks. In simpler terms, it is like a bridge. Its function is to connect the device in one network to interact with a device in another network whereas



Topic- Network Device – Difference between GATEWAY & ROUTER

Difference between Gateways and Router

Gateways	Router
A gateway is a device that is used for communication between networks having different sets of protocols.	Route is a device that receives, analyzes, and forwards the data packets to other networks.
Gateway connects two different networks.	It routes the data packets via the same networks.
Gateway does not support dynamic routing.	Router supports the dynamic routing.
The main function of a gateway is protocol translation.	The main function of a router is routing the traffic from one network to another network.
It is hosted on dedicated applications, physical servers, or virtual applications.	It is hosted on only the dedicated applications.
The gateway is also called a gateway router, proxy server, and voice gateway.	The router is also called a wireless router and an Internet router.

Topic- Network Device – Difference between GATEWAY & MODEM

Basis of Comparison	Gateway	Modem
Definition	It is a device that is the center point for two or more networks to communicate with each other.	A device that connects a home or office device to the internet.
Function	Any device that enables traffic flow in and out of the network can function as a gateway.	It converts the digital signals to analog to transmit them in cable lines later converting them back to digital at the receiver.
Integration	It includes a router that directs traffic between devices on a network.	Does not include a router instead mainly on the connection of devices to the internet.
Protocols	May include protocols like a firewall, Wi-Fi access point, or Ethernet switch.	Supports one or two protocols such as cable or DSL.
Uses	Commonly used in businesses and large organizations.	Commonly used in homes.
Connection	A gateway connects to multiple devices.	A modem connects to a single device.
IP address management	A gateway can support multiple private IP addresses of multiple devices	A modem supports a single public IP address.
Cost	Gateway is expensive.	The modem is cheaper than Gateway.

THANK YOU